MILITARY SPECIFICATION

IGNITER

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of igniter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" by 48" Pallets, Skids, Runners or Pallet-Type Base.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC 1365

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS AND TECHNICAL DATA PACKAGE (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 36-2-104 - Igniter.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 36-2-104 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the igniter shall conform to the specifications and drawings listed on TDPL 32-3-104 and on the subsidiary drawings.

3.2 Assembly. The igniter shall be assembled as shown on C36-2-104.

3.3 Filling.

3.3.1 Delay mixture. The delay mixture shall conform to C143-2-5.

3.3.2 First fire mixture. The first fire mixture shall conform to B143-9-2.

3.4 Loading. The igniter shall be loaded with a delay mixture with a first fire pellet placed at each end in such a manner as to insure a hard adhesive plug (see 6.3).

3.5 Functioning. The igniter shall ignite and burn to completion for not less than 10 nor more than 20 seconds when tested in accordance with 4.4.4.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot of igniters shall be produced for examination and tests in accordance with this specification.

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3.7 Workmanship. The igniter shall be free from damage, cracks, sharp edges and foreign matter (oil and grease) that would impair its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3).
(b) Quality conformance inspection (4.4).

4.3 Preproduction inspection.

4.3.1 Lot. Two hundred and fifty (250) igniters shall be produced using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Fifty (50) igniters shall be examined in accordance with the classification of defects. The AQL's do not apply.

4.3.2.2 For destructive test. Eight igniters shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with 4.3.2.1 and 4.3.2.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of igniters shall be identified and controlled in accordance with MIL-STD-1167 and with MIL-STD-1168, produced by one manufacturer from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be conducted on the S-4 level of MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The igniter shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive test. Each sample igniter shall be tested in accordance with 4.4.4, using an AQL of 1% in MIL-STD-105.

4.4.3.3 Classification of defects.

(a) Igniter (Drawing C36-2-104).

Categories	Defects
Critical:	None defined
Major:	AQL 0.65 percent defective
101	Filling missing, separated or cavitated
102	Thread missing or incorrect (Dwg C36-2-106)
103	First fire mixture missing or inadequate
	AQL 1.0 percent defective
104	Inside diameter incorrect
105	Chamfer missing
106	Length incorrect
107	Outside diameter incorrect
108	Workmanship (3.7)

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4.4.4 Functioning test. The igniter shall be assembled to a fuze adapter (see 6.4) in an upright position with the threaded end down inside a fuze adapter so that the upper surface of the tube is not less than 1/2 inch from the bottom of the fuze. A fuze shall be fired and the burning time from the spit of the fuze to the ejection of the lower plug of the igniter checked by means of a stop watch. If failure of the igniter to fire is observed to be the result of the fuze, a second fuze may be utilized.

5. PACKAGING

5.1 Interplant shipment. A designated quantity of igniters shall be packaged and packed to prevent damage in shipment from the supply source to the first receiving activity or immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Palletization. When specified in the contract or order, igniters shall be palletized in accordance with MIL-STD-147.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123 and shall include the contract or purchase order number and name and address of the prime contractor. Palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use in assembling the M4A2 floating smoke pot.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether to proceed with production.

6.3 Loading. A minimum of 18 grams of delay mixture is required to obtain a satisfactory delay time. 1.00 $\pm$ .015 grams of first fire mix can be placed in each end of the tube. Both first fire mix and delay should be pelletized prior to assembly and reconsolidated in the tube in a manner to provide a hard adhesive plug with no separating between pellets.

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6.4 Test accessories. The 207A1 floating smoke pot fuze conforming to MIL-F-51466 and the fuze adapter conforming to Drawing D36-2-67 may be obtained from the contracting officer on request.

6.5 Precaution. This specification covers sampling and testing of chemical toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A068

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-51466

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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